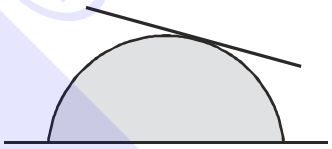
SECTION-I

Single Correct Answer Type 2 Q. [3 M (–1)]

1. A particle of mass 1 kg, executing simple harmonic motion of period 20 s, crosses the mean position at t = 0 with velocity π cm/s.
(A) the maximum acceleration of the particle is 10 cm/s^2
(B) when it is 4 cm from the mean position, its velocity is 6π/10 cm/s
(C) the kinetic energy of the particle when its displacement is 5 cm from the mean position is $\frac{3\pi^2}{8\times10^6}$ J
(D) its velocity at displacement 6 cm from the mean position is 8π/10 cm/s

2. A uniform rod of length 2ℓ rocks to and fro on the top of a rough semicircular fixed cylinder of radius a. The period of small oscillations, of rod is

(A) $\frac{2\pi\ell}{\sqrt{3ga}}$ (B) $2\pi\sqrt{\frac{\ell}{3g}}$

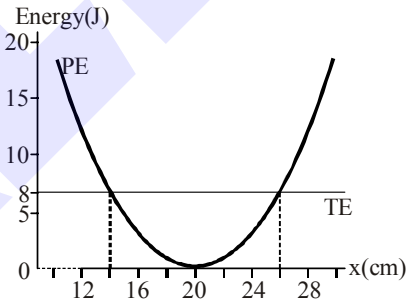


(C) $2\pi\sqrt{\frac{a}{3g}}$ (D) $\frac{\pi\ell}{\sqrt{3ga}}$

Multiple Correct Answer Type 17 Q. [4 M (–1)]

3. A particle is performing SHM with its position given as x = 2 + 5 sin $\left(\pi t+\frac{\pi}{6}\right)$ where x (in m) & t (in sec). Which of the following is/are correct :-
(A) Equilibrium position is at x = 2m
(B) Maximum speed of particle is 5π m/s
(C) At t = 0 particle is 2.5 m away from mean position moving in negative direction
(D) At t = 0, x = 4.5 m; acceleration of particle is $-\pi^2(4.5)$ m/s^2

4. Figure shows the potential-energy diagram and the total energy line of a particle oscillating on a spring.



(A) the amplitude of oscillation of particle is 6 cm.
(B) the spring constant is 4.44 kN/m.
(C) the mass of particle is 1kg.
(D) the maximum kinetic energy of particle is at x = 20 cm.

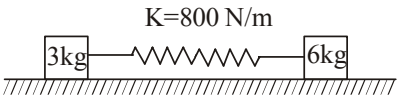
5. A linear harmonic oscillator of force constant 2×10^6Nm^{-1} and amplitude 0.01 m has a total mechanical energy of 160 J. Its
(A) maximum potential energy is 100 J (B) maximum kinetic energy is 100 J
(C) maximum potential energy is 160 J (D) minimum potential energy is zero.

6. A body is performing S.H.M., then its
(A) average total energy of SHM per cycle is equal to its maximum kinetic energy.
(B) average kinetic energy per cycle is equal to half of its maximum kinetic energy.
(C) mean velocity over a complete cycle is equal to 2/π times of its maximum velocity.
(D) root mean square velocity is 1/√2 times of its maximum velocity

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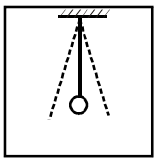
7. The potential energy of a particle of mass 0.1 kg, moving along the x-axis, is given by U = 5x (x – 4)J, where x is in metres. It can be concluded that :
(A) the particle is acted upon by a constant force.
(B) the speed of the particle is maximum at x = 2 m.
(C) the particle executes simple harmonic motion
(D) the period of oscillation of the particle is π/5 s.

8. The system shown in the figure can move on a smooth surface. The spring is initially compressed by 6 cm and then released .



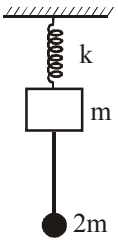
(A) the particles perform SHM with time period $\frac{\pi}{10}$ sec
(B) the block of mass 3 kg perform SHM with amplitude 4 cm
(C) the block of mass 6 kg will have maximum momentum 2.40 kg m/s
(D)none of these

9. A simple pendulum is oscillating in a stationary lift with a time period T and an angular amplitude θ_0. When it is passing through the mean position, the lift starts accelerating upwards.



(A) The time period of its oscillation will increase
(B) Its angular amplitude would increase.
(C) The time period of its oscillation would decrease
(D) Its angular amplitude would decrease.

10. A bob of mass 2m hangs by a string attached to the block of mass m of a spring block system. The whole arrangement is in a state of equilibrium. The bob of mass 2m is pulled down slowly by a distance x_0 and released.

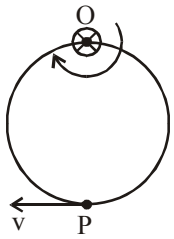


(A) For $x_0 = \frac{3mg}{k}$ maximum tension in string is 4mg
(B) For $x_0 > \frac{3mg}{k}$, minimum tension in string is mg
(C) Frequency of oscillation of system is $\frac{1}{2\pi}\sqrt{\frac{k}{3m}}$, for all non-zero values of x_0
(D) The motion will remain simple harmonic for $x_0 \le \frac{3mg}{k}$

11. Consider a spring that exerts the following restoring force
$F = -kx$ for $x > 0$
$F = -4kx$ for $x < 0$
A mass m on a frictionless surface is attached to the spring displaced to x = A by stretching the spring and released :
(A) The period of motion will be $T = \frac{3}{2}\pi\sqrt{\frac{m}{k}}$
(B) the most negative value the mass m can reach will be $x = -\frac{A}{2}$
(C) The time taken to move from x = A to $x = -\frac{A}{\sqrt{2}}$, straight away will be equal to $\frac{5\pi}{8}\sqrt{\frac{m}{k}}$
(D) The total energy of oscillations will be $\frac{5}{2}kA^2$

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12. A ring of mass m, radius r can oscillate in a vertical plane about is top most point 'O' as shown. Axis is normal to the plane of ring. Maximum speed of lowest point 'P' of ring is v.



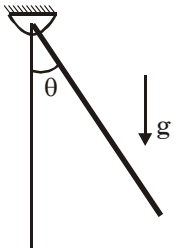
(A) Time period of SHM for small amplitudes is $2\pi\sqrt{\frac{2r}{g}}$

(B) Acceleration of centre of mass of ring at lowest position is zero.

(C) Hinge force when centre of mass is at lowest point is $\frac{mv^2}{4r}$

(D) Mechanical energy of oscillation is $\frac{mv^2}{4}$ (when potential energy at mean position is taken as zero.)

13. An uniform rod of mass m and length ℓ is pivoted at top and it can perform angular SHM in vertical plane according to equation $\theta = \theta_0 \sin \omega t$. At time $t = \frac{2\pi}{\omega}$ a point mass m (at rest) sticks to lowest end of rod.



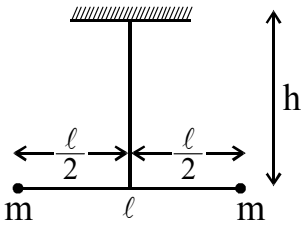
(A) New time period of rod will be $\frac{2}{\sqrt{3}}$ times of the time period before collision.

(B) New angular amplitude is $\frac{\theta_0}{\sqrt{12}}$

(C) New angular amplitude is $\frac{\theta_0}{4}$

(D) There will always be loss of kinetic energy of rod, irrespective of time at which mass sticks to rod.

14. Figure shows two particles connected with a light rod and entire arrangement hangs on a elastic wire. Neglect any gravitational effect, θ_0 = maximum angular displacement of the particles, C = torsional stiffness of wire. Mark the **CORRECT** statements :-



(A) Time period of oscillation is $\pi\ell\sqrt{\frac{2m}{C}}$

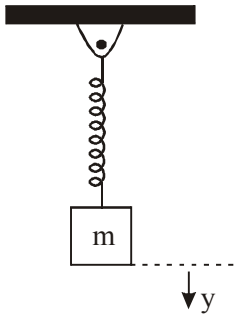
(B) Maximum tension in the rod is $\frac{C\theta_0^2}{\ell}$

(C) The energy of oscillation is $\frac{1}{2}C\theta_0^2$

(D) The maximum tangential acceleration of particle is $\frac{C\theta_0}{m\ell}$

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15. A block suspended from a spring at natural length and is free to move vertically in the y-direction. Mark the **CORRECT** statement (s) :-



(A) Mass is released when y = 0; the maximum value of y reached by the mass m is 2mg/k

(B) Mass is released when y = 0; the maximum value of y reached by the mass m is mg/k

(C) Due to air resistance the mass settles down into an equilibrium position y_{eq}, the mechanical energy loss is $\frac{1}{2}(m^2g^2/k)$

(D) Due to air resistance the mass settles down into an equilibrium position y_{eq}, the mechanical energy lost is (m^2g^2/k)

16. A particle performing S.H.M. undergoes displacement of A/2 (where A = amplitude of S.H.M.) in one second. At t = 0 the particle was located at either extreme position or mean position. The time period of S.H.M. can be : (consider all possible cases) :-

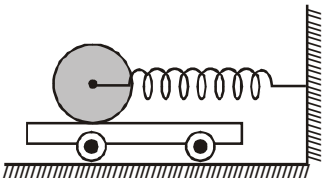
(A) 12 s (B) 2.4 s (C) 6s (D) 1.2 s

17. Two particles A and B are performing SHM along same straight line with same amplitude a and same angular frequency ω. At t = 0, A is left extreme and B is at a distance a/2 from mean position to the right and moving toward mean position. Then:

(A) their phase difference is $\frac{5\pi}{3}$ (B) their phase difference is $\frac{2\pi}{3}$

(C) time after which they cross each other is $\frac{2\pi}{3\omega}$ (D) time after which they cross each other is $\frac{\pi}{3\omega}$

18. If the figure shown, the mass of the disc as well as that of the trolley is M. The spring is ideal and has stiffness k. The trolley can move horizontally on the floor without friction, and the disc can roll on the trolley surface without slipping. The spring is compressed and the system released so that oscillations begin. At the instant of release, magntiude of :



(A) Acceleration of center of disc = twice of that of trolley

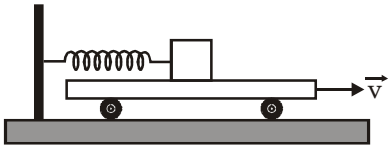
(B) Acceleration of center of disc = thrice of that of trolley

(C) Spring force is four times of friction.

(D) Spring force is two times of friction.

19. A block of mass m is attached to a fixed support by a horizontal spring with force constant k and negligible mass as shown in figure. Hooke's law describes the spring both in extension and in compression. The block placed on a very long horizontal board, with which it has coefficient of static friction μ_s and a smaller coefficient of kinetic friction μ_k. The board moves to the right at constant speed v. Assume the block spends most of its time sticking to the board and moving to the right, so the speed v is very small in comparison to the average speed the block has as it slips back toward the left. Ignore the time when the speed of block changes from v to 0 and from 0 to v. Mark the **CORRECT** statement(s):

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(A) The maximum extension of the spring from its unstretched position is very nearly given by $\frac{\mu_s mg}{k}$

(B) The block oscillates about an equilibrium position at which the spring is stretched by $\frac{\mu_k mg}{k}$

(C) The amplitude of the block's motion is $A = \frac{(\mu_s - \mu_k)mg}{k}$

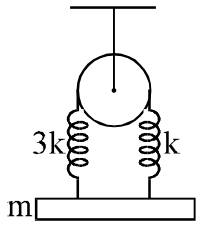
(D) The period of the block's for a complete oscillation is $T = \frac{2(\mu_s - \mu_k)mg}{vk} + \pi\sqrt{\frac{m}{k}}$

SECTION-II

Numerical Answer Type Question **1Q.[3M (0)]**

(upto second decimal place)

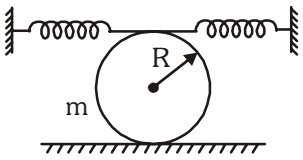
20. A uniform rod of mass m is suspended on two springs at equal distance from ends as shown. The force constants of the springs are k and 3k. During the vertical oscillations of the system, the rod remains horizontal. Find the angular frequency (in rad./s) of these osicllations. The strings and the pulley are ideal. (Given m = 3 kg, k = 676 N/m)



SECTION-III

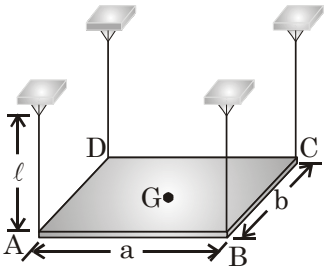
Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

21. A solid uniform cylinder of mass m performs small oscillations due to the action of two springs combined stiffness equal to k (figure). The period of these oscillations in the absence of sliding is $T = \pi\sqrt{\frac{xm}{2k}}$. Then find x :



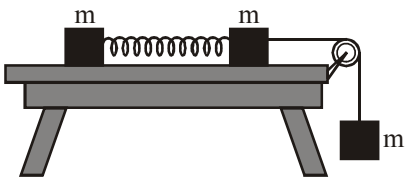
22. A car accelerates uniformly from rest. Initially, its door is slightly ajar (open). Calculate how far (in m) the car travels before the door slams shut. Assume the door has a frictionless hinge, a uniform mass distribution and a length of 1.2 m from front to back.

23. A thin rectangular plate of mass M, sides a and b is suspended from four vertical wires of the same length ℓ. Determine the period of small oscillations of the plate when it is given a small horizontal displacement in a direction perpendicular to AB for a = b = ℓ. If your answer is $T = 2\pi\sqrt{\frac{x\ell}{yg}}$ fill value of x + y.

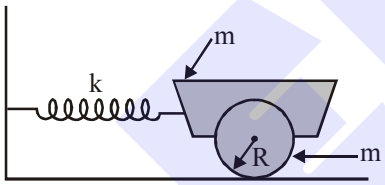


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24. At a smooth horizontal table are two identical cubes of mass m, connected by a spring of rigidity k. The length of spring in the unstretched state is ℓ_0. The right cube is linked to the load mass m at the end. At some time the system is released and the system moves without initial velocity. Find the maximum distance (in cm) between blocks during the motion of the system.
[ℓ_0 = 1 cm, m = 3 kg, k = 1000 N/m]



25. A cart consists of a body of mass m and two wheels, each of mass m and radius R. The cart is attached to a spring of constant k. The other end of the spring is fixed to a wall as shown in figure. If time period of oscillation is $\beta \times 2\pi\sqrt{\frac{m}{k}}$ then find the value of β



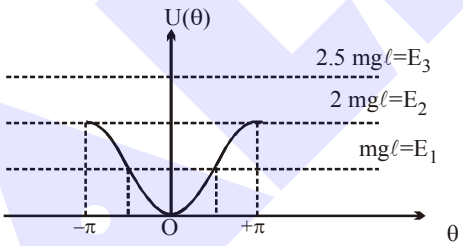
SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0))]**

26. Column-I represents potential energy graph for certain system. Column-II gives statements related to graphs.

Column-I **Column-II**

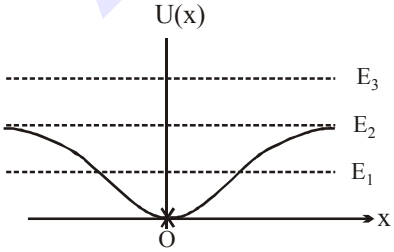
(A)



U vs θ graph for a bob hanging vertically from a string with its lowest position as reference level and θ is angle of string from vertical line

(P) If total energy is E_3, it is not possible for the body to have any turning point in its motion.

(B)

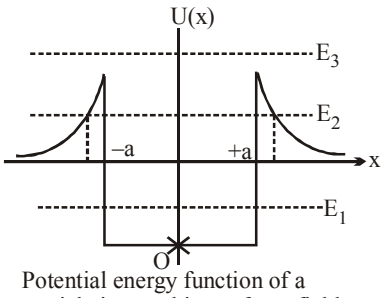


A particle moving along x axis with potential energy function as U(x) = $[1-e^{-x^2}]$

(Q) For a small displacement about point O potential energy function is quadratic in variable plotted on x-axis.

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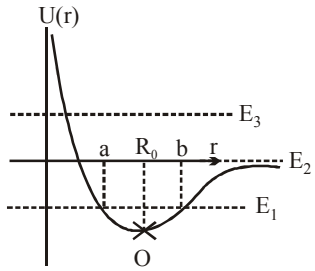
(C)



Potential energy function of a particle in an arbitrary force field

(R) For a small displacement about position O motion is simple harmonic.

(D)



Graph represents potential energy for a particle

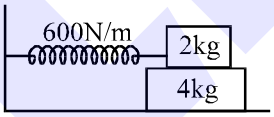
(S) If total energy is $E_{total} < E_2$ particle executes periodic and oscillatory motion for all energy values greater than energy at O.

(T) Point O is position of stable equilibrium

Subjective Type

4 Q. [4 M (0)]

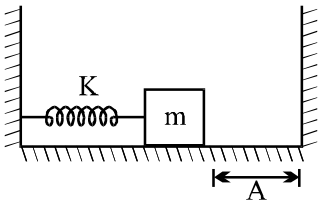
27. In the arrangement shown, the spring of force constant 600N/m is in the unstretched position. The coefficient of friction between the two blocks is 0.4 and that between the lower block & ground surface is zero. If both the blocks are displaced slightly and released, the system executes SHM.



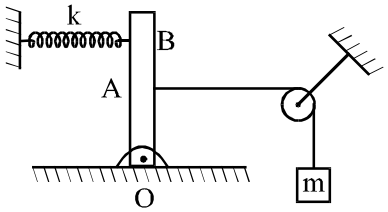
(a) Find time period of their oscillation if they do not slip w.r.t. each other.

(b) What is the maximum amplitude of the oscillation for which sliding between them does not occur.

28. In the figure shown the spring is relaxed. The spring is compressed by 2 A and released. Mass m attached with the spring collides with the wall & loses two third of its kinetic energy & returns. Find the time after which the spring will have maximum compression first time after releasing. (Neglect Friction)

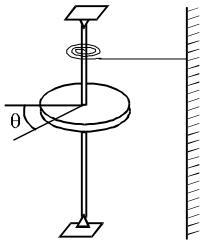


29. A massless rod is hinged at O. A string carrying a mass *m* at one end is attached to point A on the rod so that OA = a. At another point B (OB= b) of the rod , a horizontal spring of force constant *k* is attached as shown. Find the period of small vertical oscillations of mass *m* around its equilibrium position.



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30. A solid disk of mass M and radius R on a vertical shaft. The shaft is attached to a coil spring which exerts a linear restoring torque of magnitude Cθ where θ is the angle measured from the static equilibrium position and C is a constant. Neglect the mass of the shaft and the spring and assume the bearings to be frictionless.



(a) Show that the disks can undergo SHM and find the frequency of motion.
(b) Suppose that the disk is moving according to the $\theta = \theta_0 \sin(\omega t)$ where ω is the frequency found in part (a) At time $t_1 = \pi/\omega$ a ring of stickly putty of mass M and radius R is dropped concentrically on the disk, Find (i) the new frequency of motion (ii) the new amplitude of motion.

GR # SHM **ANSWER KEY**

SECTION-I

Single Correct Answer Type **2 Q. [3 M (–1)]**

1. Ans. (D) **2. Ans. (A)**

Multiple Correct Answer Type **17 Q. [4 M (–1)]**

3. Ans. (A, B) **4. Ans. (A,B,D)** **5. Ans. (B,C)** **6. Ans. (A,B,D)** **7. Ans. (B,C,D)**
8. Ans. (A,B,C) **9. Ans. (C,D)** **10. Ans. (A,D)** **11. Ans. (AB)** **12. Ans. (A, D)**
13. Ans. (A, B) **14. Ans. (A, B, C, D)** **15. Ans. (A, C)** **16. Ans. (A,B,C,D)**
17. Ans. (B,D) **18. Ans. (B,C)** **19. Ans. (A,B,C,D)**

SECTION-II

Numerical Answer Type Question (upto second decimal place) **1Q.[3M (0)]**

20. Ans. 26.00

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

21. Ans. 3 **22. Ans. 1** **23. Ans. 2** **24. Ans. 3** **25. Ans. 2**

SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

26. Ans. (A)-P,Q,R,T; (B)-P,Q,R,S,T; (C) P; (D)-Q,R,S,T

Subjective Type **4 Q. [4 M (0)]**

27. Ans. (a) π/5, (b) 2cm **28. Ans.** $\frac{17\pi}{12}\sqrt{\frac{m}{K}}$ **29. Ans.** $(2\pi a/b)(m/k)^{1/2}$

30. Ans. (a) $\frac{1}{2\pi}\sqrt{\frac{2C}{MR^2}}$, (b) $\theta_{new} = \frac{\theta_0}{\sqrt{3}}, f' = \frac{1}{2\pi}\sqrt{\frac{2C}{3MR^2}}$

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